

PHY 111 (Spring 2025)  
PRINCIPLES OF PHYSICS I  
**Assignment-1**

***Due date: 12<sup>th</sup> March 2025.***

1. Two cannon balls A and B are projected at the same time from point O in the different directions with different velocities 35 m/s and 39 m/s respectively, which are creates some angles with the Cartesian coordinate as shown in the Figure-1. Both the balls strike the ground in different location and embedded into the ground. Neglect the air friction for the following calculations.
- (i) (2 marks) Calculate the flight of time for each of the both balls.
  - (ii) (4 marks) Calculate the velocities of the both ball at  $t = 2$  sec.
  - (iii) (4 marks) Determine the displacement between the balls after being striking the Ground.

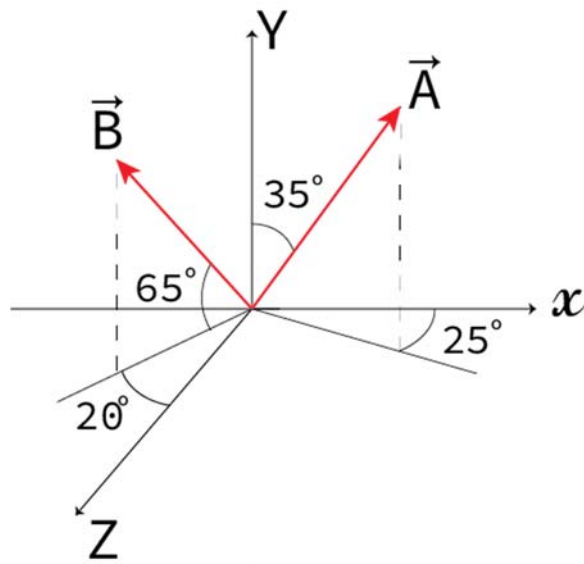


Figure-1